

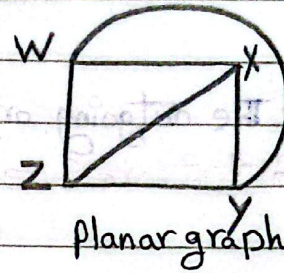
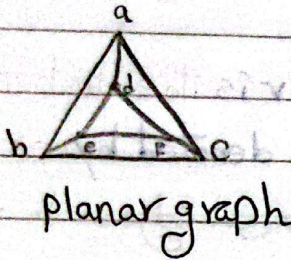
CH: 11

Planar Graph

Lecture 8

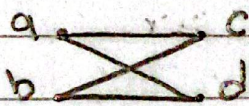
* Def: A graph or multigraph G is called **planar** if G can be drawn in the plane with its edges intersecting only at vertices of G .

Ex:



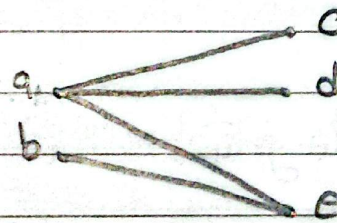
* Def: A graph $G = (V, E)$ is called **bipartite** if $V = V_1 \cup V_2$ with $V_1 \cap V_2 = \emptyset$ and every edge of G is of the form $\{a, b\}$ with $a \in V_1$ and $b \in V_2$, if each vertex in V_1 is joined with every vertex in V_2 , then we have **complete bipartite graph**, in this case if $|V_1| = m$ and $|V_2| = n$, then the graph is denoted by $K_{m,n}$.

Ex:



$$V_1 = \{a, b\}$$

$$V_2 = \{c, d\}$$

complete bipartite ($K_{2,2}$)

$$V_1 = \{a, b\}$$

$$V_2 = \{c, d, e\}$$

bipartite graph
but not complete

* Def: Let $G = (V, E)$ be a loop-free undirected graph, where $E \neq \emptyset$. An **elementary subdivision** of G results when edge $e = \{u, w\}$ is removed from G and then the edges $\{u, v\}$, $\{v, w\}$ are added to $G - e$, where $v \notin V$.

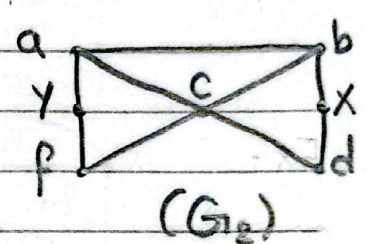
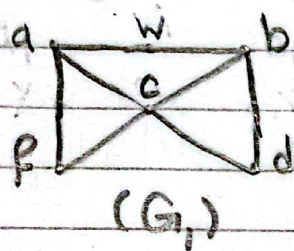
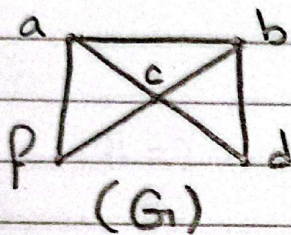
*

The Loop-free undirected graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are called homomorphic if

① they are isomorphic

② they can both be obtained from the same loop-free undirected graph H by a sequence of elementary subdivisions

Ex:



- G_1 is obtained from G by one elementary subdivision, delete edge $\{a, b\}$ from G and add the edges $\{a, w\}$ and $\{w, b\}$
- the graph G_2 is obtained from G by two elementary subdivision
- G_1, G_2 are homomorphic

* theorem: (Kuratowski's theorem)

⑤

A graph is nonplanar iff it contain a subgraph that is homomorphic to K_5 or $K_{3,3}$

يعني لو قمت Subgraph من ال graph الاساسي وكان مشابه لـ K_5 أو $K_{3,3}$ يبقى هو nonplanar

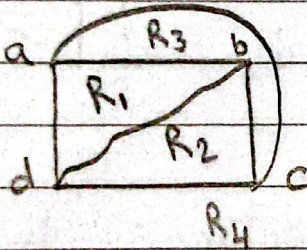
* theorem: (Euler's theorem)

⑥

Let $G = (V, E)$ be a connected Planar graph or multigraph with $|V| = v$ and $|E| = e$, Let r be the number of region in the plane determined by a planar embedding (or depiction) of G , one of these regions has infinite area and is called infinite region, then $v - e + r = 2$

Proof: موجود في الكتاب صفحة 545

Ex:



$$|V| = 4 \quad |E| = 6 \quad r = 4$$

$$\therefore 4 - 6 + 4 = 2, R_4 \rightarrow \text{infinite region}$$

* Corollary: Let $G = (V, E)$ be a loop-free connected Planar graph with $|V| = v$ and $|E| = e > 2$ and r regions, then $3r \leq 2e$ and $e \leq 3v - 6$

③

Proof: موجود في الكتاب صفحة 547

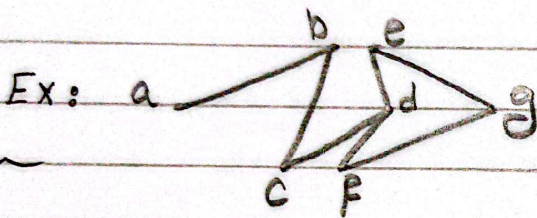
Ex: The graph K_5 is a Loop-free and connected with $|E| = 10, |V| = 5$

$$\therefore 3v - 6 = 3 \times 5 - 6 = 9 < 10 = e \therefore 3v - 6 < e$$

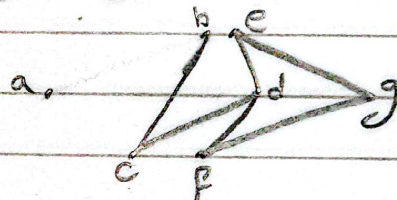
$\therefore$ by corollary ③ K_5 is non planar

* Def: Let $G = (V, E)$ be an undirected graph or multigraph, A subset E' is called a cut-set of G if by removing the edges (but not the vertices) in E' from G , we have $K(G) < K(G')$, where $G' = (V, E - E')$, but when we remove (from E) any proper subset E'' of E' , we have $K(G) = K(G'')$, for $G'' = (V, E - E'')$

بأختصار الكلام ده معناه ان ال cut-set دى عبارة عن set فيما ال edges اللي لو اتشالت، تهمل 2 components من حاجة من اتنين ① اتنين graph منفصلين عن بعض (Subgraphs) ② isolated vertex و subgraph

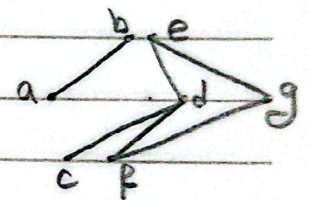


(G)



(G')

cut-set: $\{\{a, b\}\} = E'$
 $\{a, b\}$ is a bridge
 a isolated



(G'')

cut-set: $\{\{b, c\}\}$
 $\{b, c\}$ is a bridge
 there are 2 component

Notes:

① Let $m, n \in \mathbb{Z}^+$, $m \geq n \geq 2$

$\therefore$ the no. of cycles of Length 4 in $K_{m,n}$ is $C_2^m \times C_2^n$

$\therefore$ the no. of different Paths of Length 2 in $K_{m,n}$ is $[{}_m C_2^n + n C_2^m]$